Drop2Crop — System User Manual

Smart Dual-Motor IoT Precision Irrigation & Fertigation Station

Target Device: ESP32-S3 (16MB Flash, 8MB PSRAM)	**Firmware Version:** v2.5-DualMotor-Production
Cloud Protocol: MQTT over Secure WebSockets (WSS)	**Broker:** test.mosquitto.org (Port 8081 / 1883)
Dual Actuators: Motor 1: Clean Water (Relay 1) | Motor 2: Fertigation (Relay 2)	**Timezone:** Asia/Dhaka (BST, UTC+6)
Dashboard URL: https://fahimashabb.github.io/drop2crop/	**Security Gateway:** 4-Digit Hardware-Verified PIN (Default: 2564)

1. System Overview & Dual-Motor Architecture

Drop2Crop is an advanced precision agricultural IoT station designed to deliver autonomous, data-driven crop irrigation and nutrient delivery without manual guesswork. The system features an industrial-grade ESP32-S3 microcontroller coupled to a 2-channel isolated relay board that operates two independent 12V pump circuits:

- **Motor 1 — Clean Water Pump (Relay 1 / GPIO 9):** Draws fresh water from the main reservoir for normal hydration, moisture maintenance, and soil flushing.
- **Motor 2 — Fertigation Pump (Relay 2 / GPIO 10):** Dedicated pump drawing liquid fertilizer, nutrient mix, or mineral additives from a separate fertilizer tank, preventing chemical fouling of the primary clean reservoir.
- **Independent & Coordinated Control:** Either pump can be run individually or both can be run simultaneously via the cloud web app hosted on GitHub Pages from anywhere across Bangladesh.

2. Zero-Code WiFi Provisioning & Relocation (Captive Portal)

Drop2Crop incorporates an intelligent **Captive Portal Provisioning System**. If you move the hardware to a new farm, switch routers, or use mobile hotspots, **you never need to plug the ESP32 into a computer or touch firmware code**:

- **Automatic Activation:** On boot, the ESP32 attempts to connect to saved WiFi. If disconnected for 12 seconds, it automatically broadcasts an open network: **Drop2Crop_WiFi_Setup**.
- **Manual Force Activation:** Hold the physical **BOOT button (GPIO 0)** on the ESP32 for 3 seconds while powered. The onboard LED will flash rapidly and launch the portal immediately.
- **Connect:** Open WiFi settings on your phone or PC, select **Drop2Crop_WiFi_Setup**.
- **Portal Page:** The browser automatically opens **http://192.168.4.1**. Select your 2.4 GHz WiFi network, enter the password, and click **Save & Connect**.
- **Instant Cloud Reconnection:** Credentials save to NVS flash, the ESP32 connects to the cloud, synchronizes Bangladesh Standard Time (UTC+6) via NTP, and resumes all operations.

3. Operational Modes & Multi-Motor Logic

Mode	Mode Name	Dual-Motor Behavior & Safeguards
Mode 0	**Smart Auto (Soil + Rain)**	Autonomous closed-loop irrigation. Turns **Motor 1 (Clean Water)** ON when soil moisture drops below **35%**, and shuts OFF at **70%** saturation. Both motors are automatically **inhibited for 24 hours** if rain occurs, and cut off if water level is below **15%**.
Mode 1	**Alert-Only Monitor**	Passive sensing. When soil drops below 35%, triggers audible cloud dashboard alarms and visual status warnings without running pumps. Farmer reviews condition and starts Water, Fertigation, or Both via manual buttons.
Mode 2	**Multi-Timer Schedule**	Clock-driven schedule. Each timer slot specifies target: ■ **Water Only**, ■ **Fertigation Only**, or ■ **Both Motors**. Executes at programmed times (e.g. 6:00 AM Water 20m, 6:00 PM Fertigation 15m).
Mode 3	**Manual Cloud Override**	Direct farmer control. Four one-touch buttons: *'Start Water Pump'*, *'Start Fert Pump'*, *'Start Both Motors'*, and *'Stop All Motors'*. Dry-run reservoir cutoff remains strictly active.

4. Precision Scheduling with Actuator Targeting

Drop2Crop's NTP hardware scheduler stores up to **10 distinct daily timer slots** permanently in NVS Flash. When adding a schedule slot, the farmer selects which motor will run:

- **Target 0 — Clean Water:** Ideal for routine morning hydration (e.g. 06:00 AM, 20 mins) or deep afternoon root soaking.
- **Target 1 — Fertigation:** Ideal for precise dosage delivery (e.g. Wednesday 08:00 AM, 10 mins) from the nutrient reservoir.
- **Target 2 — Both Motors:** Synchronous run for pre-mixed flood or combined drip lines.
- **Fail-Safe Clock:** Real-time clock tracks Bangladesh Standard Time via NTP. Continues executing offline even if Internet connection drops.
- **Anti-Duplicate Lockout:** A daily latch ensures schedules only run once per calendar day and reset cleanly at midnight (00:00:00).

5. Critical Sensor Safeguards (24h Rain Delay & Ultrasonic Dry-Run)

24-Hour Rain Lockout: Raindrop detection starts an 86,400-second (24h) countdown that pauses automated & scheduled runs for both water and fertigation to prevent nutrient wash-out and waterlogging.

Ultrasonic Water Reservoir Monitoring: HC-SR04 sonar sensor measures depth from reservoir top:
- **Level < 20%:** Triggers amber dashboard warning to refill.
- **Level < 15%:** Hard emergency interlock instantly disengages both relays, preventing pump cavitation and motor burnout.

6. Device Online / Offline Status & Health Diagnostics

- **Real-Time Online/Offline Detection:** Heartbeat packets stream every 2 seconds. If power is lost or charger unplugged, dashboard alerts **'The Device is Not Online (Turned Off)'** within 6 seconds.
- **Red Power LED:** Solid red PWR LED illuminates upon 5V power connection.
- **WiFi Beacon:** If WiFi is missing, ESP32 broadcasts *'Drop2Crop_WiFi_Setup'*, proving the CPU and RF circuitry are 100% operational.
- **Relay Mechanical Clicks:** Testing 'Start Water Pump' or 'Start Fert Pump' produces audible relay clicks and lights individual red channel LEDs.

7. Access Security & Hardware Authentication

The web dashboard is protected by a 4-digit PIN lockscreen (default **2564**). Every MQTT command requires hardware authentication token verification on the ESP32-S3, preventing unauthorized remote access.